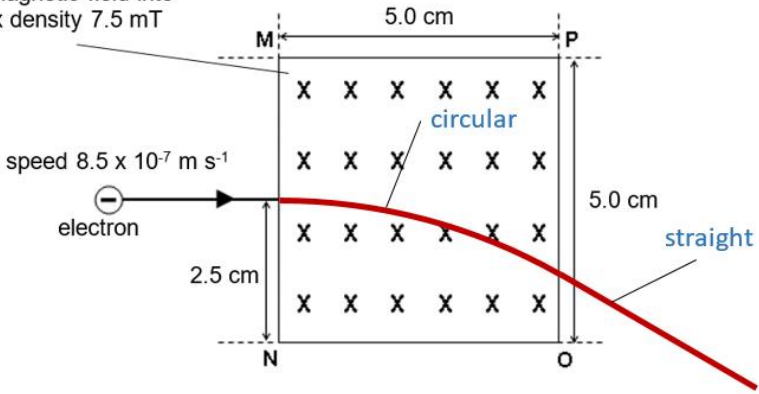


| Qn 5     | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Marks               |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| (a)      | <p>(Using Fleming's Left hand Rule), the magnetic force will <b>always be perpendicular to the direction of motion of the charge</b>. This force will provide the centripetal force for circular motion.</p> <p>Since the <b>magnitude of the magnetic force is constant</b>, the <b>magnitude of the centripetal force is constant</b>. The electron will describe a uniform circular path.</p> <p><u>Or</u> since the force is always perpendicular to the motion, no work is done by the force on the system and there is no gain in kinetic energy of the electron and hence the electron is in uniform circular motion (or moving at constant speed).</p> <p>B1 – for explaining why the motion is that of a circular one.</p> <p>B1 – for explaining why the motion is uniform</p> | <p>B1</p> <p>B1</p> |
| (b)(i)   | <p>The magnetic force provides for the centripetal force,</p> $Bqv = \frac{mv^2}{r}$ $r = \frac{mv}{Bq} = \frac{(9.11 \times 10^{-31})(8.5 \times 10^7)}{(7.5 \times 10^{-3})(1.60 \times 10^{-19})}$ $= 6.45 \times 10^{-2} \text{ m} = 6.5 \text{ cm}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <p>M1</p> <p>A1</p> |
| (b)(ii)  | <p>uniform magnetic field into page, flux density 7.5 mT</p>  <p>B1- circular path curving downwards in B-field</p> <p>B1 - emerging at side PO (allow e.c.f.)</p> <p>B1 – straight path after leaving the B-field</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | B3                  |
| (b)(iii) | <p>The magnetic force on electron provides the required centripetal force to keep the electron in circular motion.</p> <p>If <math>m</math> is the mass of the electron and <math>\omega</math> is the angular velocity of the electron.</p> $Bev = mr\omega^2 = mr\left(\frac{2\pi}{T}\right)^2 \Rightarrow T = \frac{2\pi m}{Be}$ <p>Hence, <math>T</math> is independent of <math>v</math> and <math>r</math>.</p>                                                                                                                                                                                                                                                                                                                                                                    | M1                  |
| (b)(iv)  | <p>Since period <math>T</math> is independent of the speed and radius of the path taken, the period of both electrons are the same. Hence, <u>the time spent by each electron in the magnetic field simply</u></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | B1                  |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                   |           |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
|  | <p>depends <b><u>on the fraction of the circular path traveled by each electron</u></b> within the magnetic field.</p> <p><u>Electron <math>e_1</math> travels for less than <math>\frac{1}{4}</math> of a period in the field, while electron <math>e_2</math> travels for <math>\frac{1}{2}</math> a period.</u> Hence, <b><u>electron <math>e_2</math> spends a longer time in the magnetic field.</u></b></p> | <b>A1</b> |
|  | [Total : 10 marks ]                                                                                                                                                                                                                                                                                                                                                                                               |           |